

Biochemistry Lec.3

- Monosaccharide derivatives
- Disaccharides



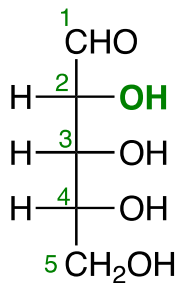
Monosaccharide derivatives

1- Sugar acids

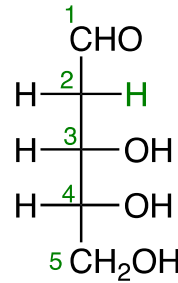
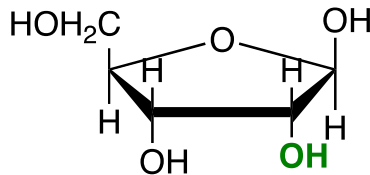
2- Sugar alcohols

3- Deoxy sugars

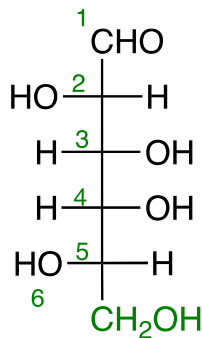
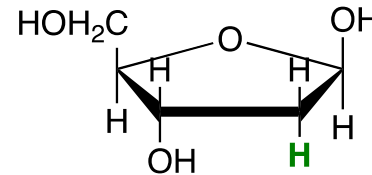
They are sugars in which one of the hydroxyl groups has been replaced by a hydrogen atom (one oxygen atom is missed).



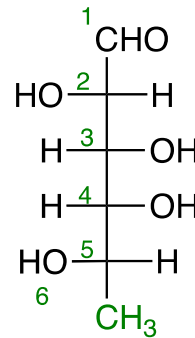
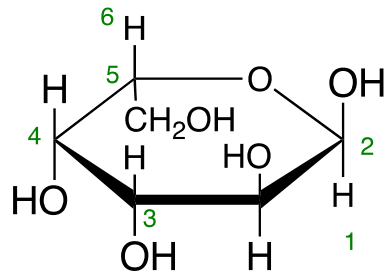
D-ribose



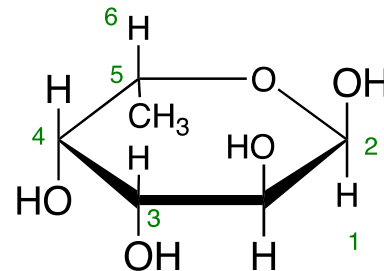
2-Deoxy-D-ribose



L-Galactose



6-Deoxy-L-Galactose
(fucose)

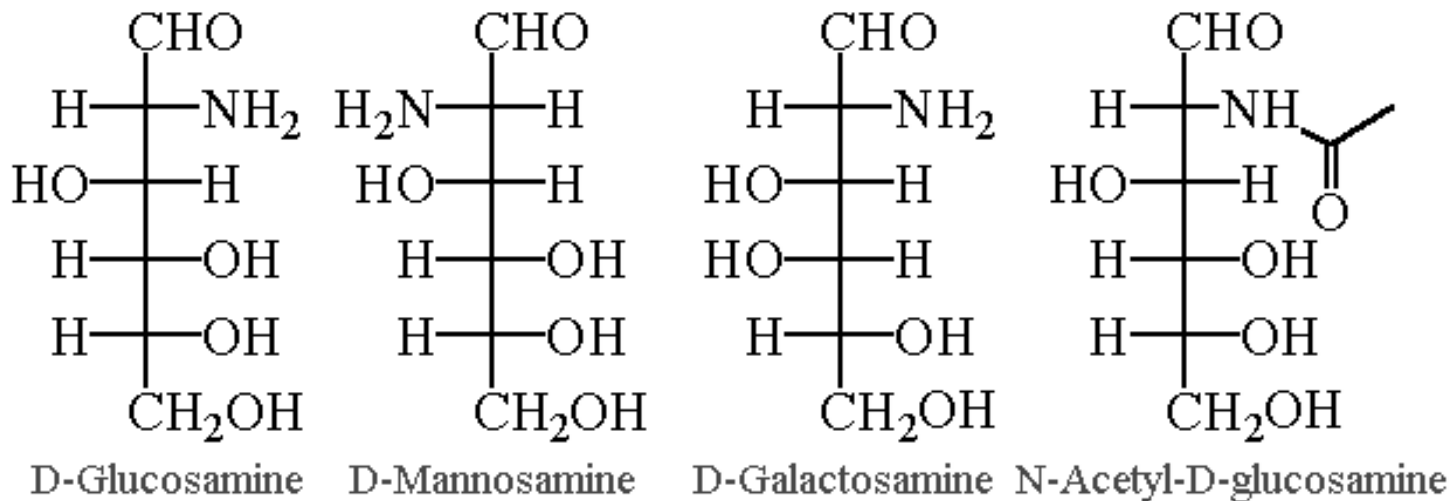


Monosaccharide derivatives

4- Amino sugars

- These sugars, the hydroxyl group is replaced by an amino or an acetylamino group.
- Amino sugars are constituents of glycoproteins, gangliosides and glycosaminoglycans.
- Glucosamine occurring in heparin and hyaluronic acid.
- Galactosamine present in chondroitin sulphate.
- Mannosamine present in neuraminic acid and sialic acid.

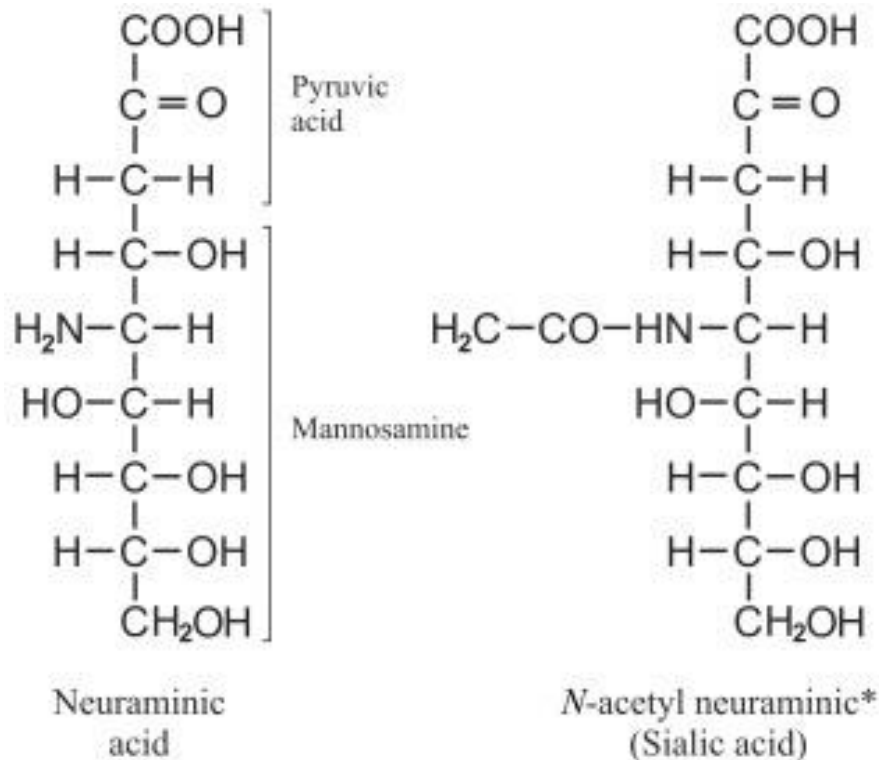
Amino Sugars



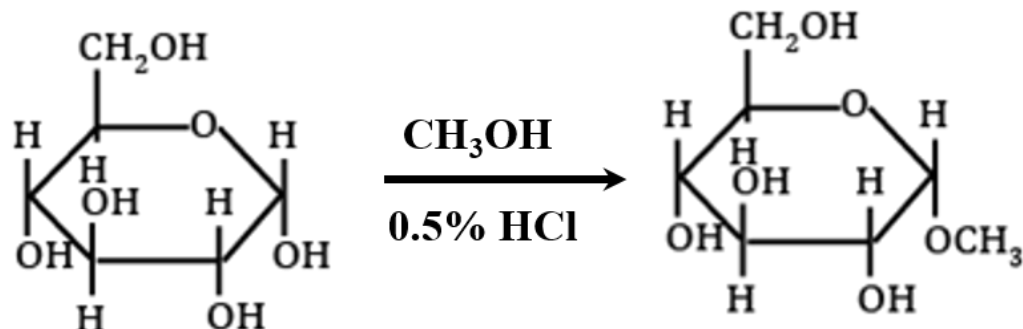
Monosaccharide derivatives

5- Amino sugar acids

- These are a condensation of amino sugars and some acids. They are occurring in glycoproteins.
- Neuraminic acid is an important building unit of the structural polysaccharides in the cell coats of higher animals composed of D-mannosamine and pyruvic acid.
- Sialic acid is N-acetyl neuraminic acid (NANA) present in glycolipids (gangliosides).



Monosaccharide derivatives

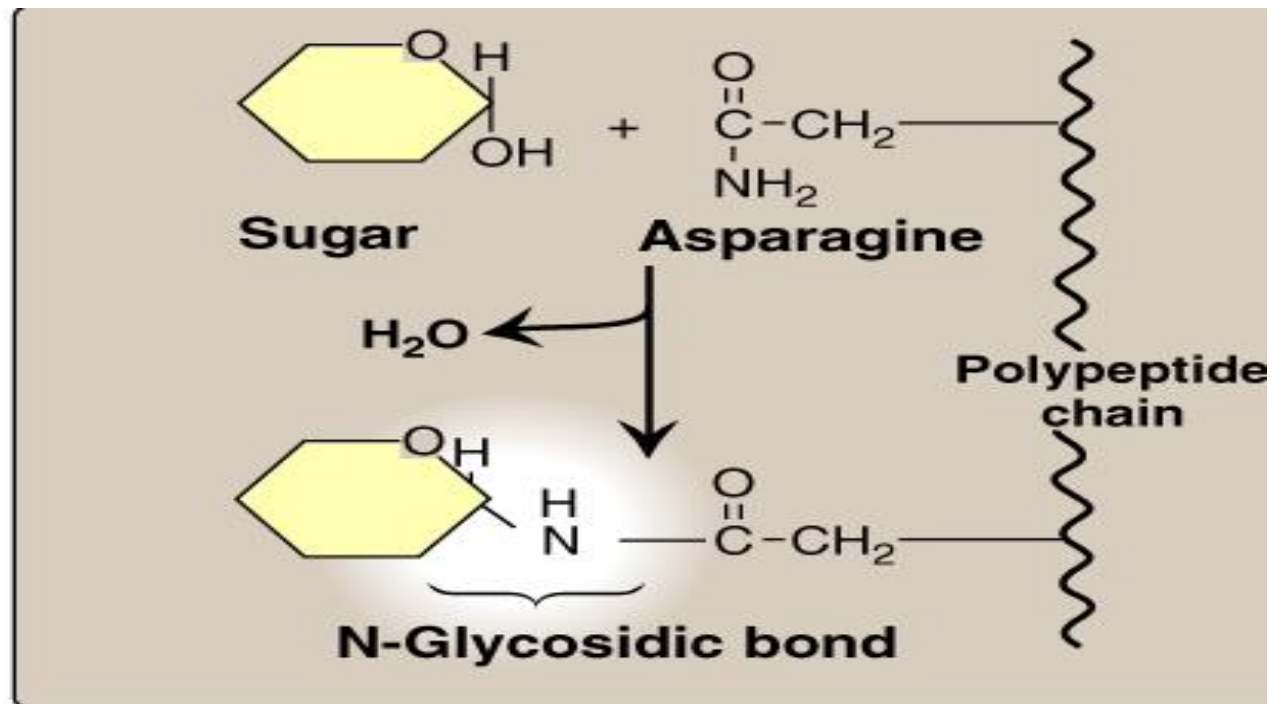


6- Glycosides

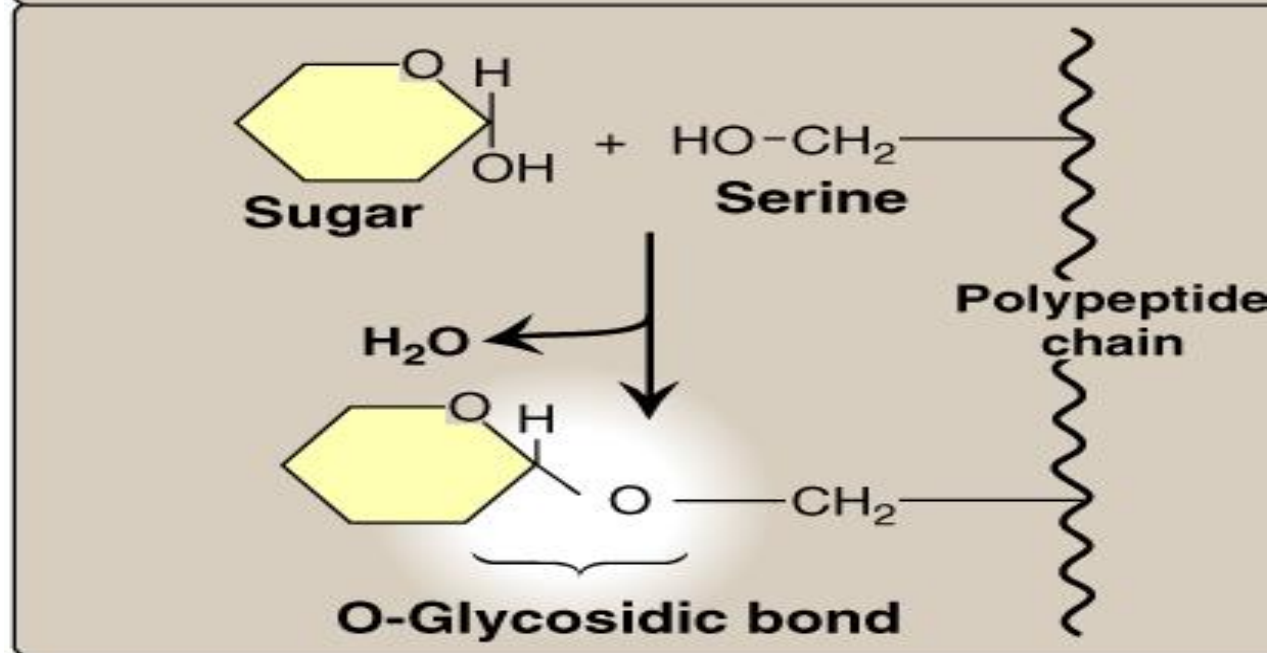
- ✓ Glycosides are derivatives of sugar.
- ✓ Glycosides has Aglycone part linked to C1 (anomeric carbon atom) of sugar by an acetal linkage (glycosidic bond).
 - Aglycone part (Non Sugar, Hydroxyl group containing compounds) e.g. Methanol, Sterol, Phenol, Glycerol.
- ✓ Glycosides has glycone part linked to C1 (anomeric carbon atom) of sugar by an acetal linkage (glycosidic bond) and so called disaccharides.
- ✓ Glycosides are naturally occurring substances present in plants and animal bodies; which are extracted and used as drugs.

S.No	Examples of Glycosides	Use
1	Cardiac Glycosides Digoxin /Digitonin Ovabain	In treatment of cardiac insufficiency.
2	Phlorizin (Glucose Transporter Inhibitor)	Induction of Diabetes mellitus.
3	Streptomycin	Antibiotic used to treat bacterial infections.
4	Glucovanillin	Flavoring agent in Ice creams and Puddings.

N-Glycosidic



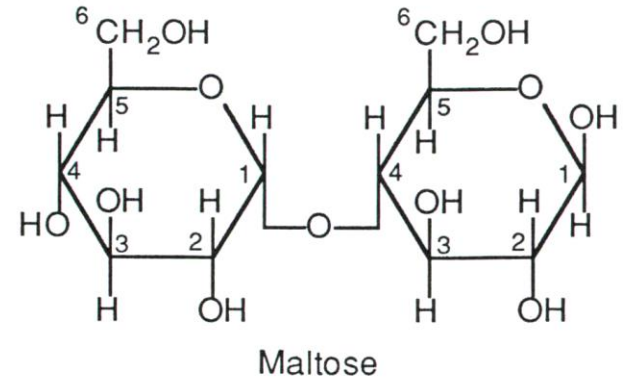
O-Glycosidic



Disaccharides

1- Maltose (Malt sugar)

- ✓ It is formed of 2 molecules of α -D glucopyranose linked together by α 1-4 glucosidic bond.
- ✓ It is a reducing sugar (can reduce Benedict reagent).
- ✓ It can be present in α and β forms.
- ✓ It can show mutarotation.
- ✓ It can form characteristic Osazone crystals.
(sunflower shape)

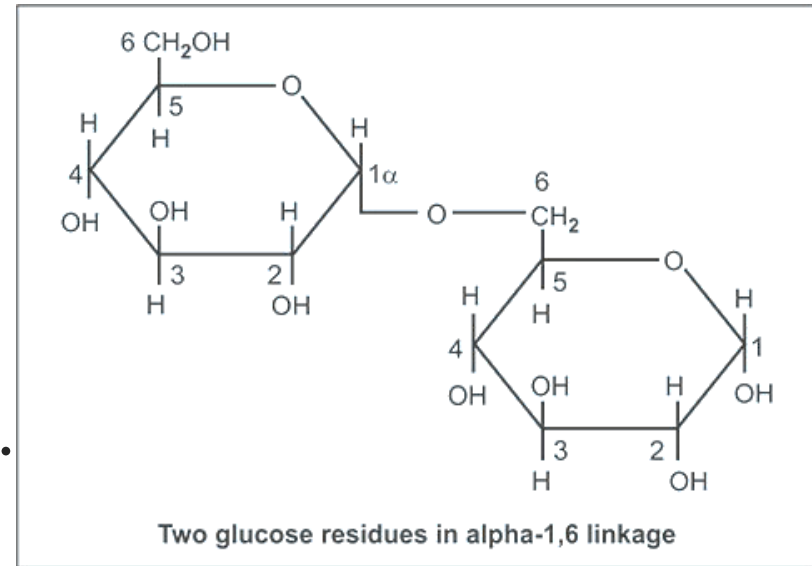


- ✓ It is fermented by yeast enzymes.
- ✓ It hydrolyzed by acids or maltase E into 2 molecule of glucose.
- ✓ It is produced by hydrolysis of starch and glycogen by amylase enzyme.

Disaccharides

2- Isomaltose

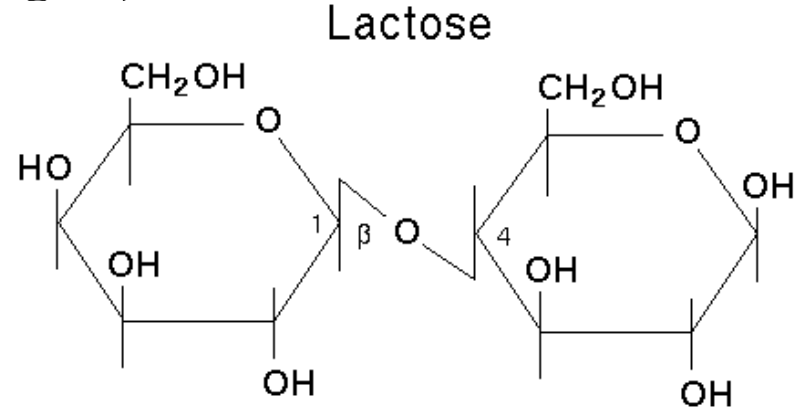
- ✓ It is formed of 2 molecules of α -D glucopyranose linked together by α 1-6 glucosidic bond.
- ✓ It is a reducing sugar.
- ✓ It can be present in α and β forms.
- ✓ It can show mutarotation.
- ✓ It can form characteristic Osazone crystals.
- ✓ It is fermented by yeast enzymes.
- ✓ It hydrolyzed by acids or isomaltase E into 2 molecule of glucose.
- ✓ It is produced by hydrolysis of starch and glycogen by amylase enzyme.



Disaccharides

3- Lactose (Milk sugar)

- ✓ It is formed of 2 molecules of β -D galactopyranose and D glucopyranose linked together by β 1-4 galactosidic bond.
- ✓ It is reducing sugar (can reduce Benedict reagent).
- ✓ It can present in α and β forms.
- ✓ It can form characteristic Osazone crystals.
(cotton ball shape)
- ✓ It can Show mutarotation.
- ✓ It cannot be fermented by yeast enzymes.



Lactose is digested by intestinal enzyme called lactase into galactose and glucose. Deficiency of this enzyme stops the digestion of lactose.

Human milk contains 7 g% of lactose while cow milk contains 4 g%

Disaccharides

4- Sucrose (Cane sugar)

- ✓ It is formed of two molecules of α -D-glucopyranose and β -D-fructofuranose linked together by α 1- β 2 glucosidic bond.
- ✓ It is not a reducing sugar.
- ✓ It cannot present in α and β forms.
- ✓ It cannot show mutarotation.
- ✓ It cannot form Osazone crystals.
- ✓ It can be fermented by yeast enzymes.

Sucrose is dextrorotatory (+ 66.5). On hydrolysis by invertase (sucrase) enzyme, it gives a mixture of equal number of glucose and fructose molecules. This mixture is called invert sugar and it is levorotatory (- 20.4).

Invert sugar used as sweetening agent.

Invert sugar used for adulteration of honey.

